

VII. SECURE NETWORK ARCHITECTURE

Aim:

Learn about and implement security best practices for network environments

Procedure:

1. Analyze the given network architecture and identify different components such as servers, clients, and security devices.
2. Understand network segmentation by separating internal, external, and DMZ networks.
3. Configure security controls like firewalls, access control lists, and intrusion detection/prevention systems.
4. Implement secure communication methods such as VPNs and encrypted protocols.
5. Apply the principle of least privilege and restrict unnecessary access between network segments.
6. Test the network for vulnerabilities and verify that security measures are properly enforced.

Tasks:

Task 1: Introduction

Read the above and continue to the next task.

No answer needed

✓ Correct Answer

Task 2: Network Segmentation

How many trunks are present in this configuration?

4

✓ Correct Answer



What is the VLAN tag ID for interface eth12?

30

✓ Correct Answer



Task 3: Common Secure Network Architecture

From the above table, what zone would a user connecting to a public web server be in?

External

✓ Correct Answer



From the above table, what zone would a public web server be in?

DMZ

✓ Correct Answer



From the above table, what zone would a core domain controller be placed in?

Restricted

✓ Correct Answer



Task 4: Network Security Policies and Control

According to the corresponding ACL policy, will the first packet result in a drop or accept?

accept

✓ Correct Answer

According to the corresponding ACL policy, will the second packet result in a drop or accept?

drop

✓ Correct Answer

Task 5: Zone-Pair Policies and Filtering

What is the flag found after filling in all blanks on the static site?

THM{M05tly_53cure}

✓ Correct Answer

Task 6: Validating Network Traffic

Does SSL inspection require a man-in-the-middle proxy? (Y/N)

Y

✓ Correct Answer

What platform processes data sent from an SSL proxy?

Unified Threat Management

✓ Correct Answer

Task 7: Addressing Common Attacks

Where does DHCP snooping store leased IP addresses from untrusted hosts?

DHCP Binding Database

✓ Correct Answer

Will a switch drop or accept a DHCPRELEASE packet?

Drop

✓ Correct Answer

Does dynamic ARP inspection use the DHCP binding database? (Y/N)

Y

✓ Correct Answer

Dynamic ARP inspection will match an IP address and what other packet detail?

MAC Address

✓ Correct Answer

Task 8: Conclusion

Read the above and continue learning!

No answer needed

✓ Correct Answer

Result:

Successfully designed and analyzed a secure network architecture by implementing proper segmentation, access controls, and security mechanisms, ensuring protection against unauthorized access and potential threats.